

Summary Of Common Errors And Solutions

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1. Multi_Brains_Pre Package
 - 1.1 Configuration File Related Errors
 - Error 1.1.1: Configuration File Not Found
 - Error 1.1.2: YAML Parsing Failed
 - 1.2 ASR Speech Recognition Related Errors
 - Error 1.2.1: ASR Engine Initialization Failed
 - Error 1.2.2: Speech Recognition Failed
 - Error 1.2.3: VAD Engine Initialization Failed
 - Error 1.2.4: Audio Stream Open Failed
 - Error 1.2.5: No Speech Detected
 - Error 1.2.6: Unsupported Language
 - 1.3 TTS Speech Synthesis Related Errors
 - Error 1.3.1: TTS Engine Initialization Failed
 - Error 1.3.2: Speech Synthesis Failed
 - 1.4 Voice Wakeup Module Serial Communication Related Errors
 - Error 1.4.1: Serial Port Open Failed
 - Error 1.4.2: Serial Port Reconnection Failed
 - Error 1.4.3: Serial Port Send Failed
 - 1.5 ROS Communication Related Errors
 - Error 1.5.1: Action Server Unavailable
 - Error 1.5.2: Action Goal Rejected
 - 1.6 Large Model Service Related Errors
 - Error 1.6.1: Dify Connection Failed
 - Error 1.6.2: Model Request Failed
 - 1.7 Motion Control Related Errors
 - Error 1.7.1: TF Transform Acquisition Failed
 - Error 1.7.2: TF Timestamp Too Old
 - Error 1.7.3: Joint Values Not Initialized
 - 1.8 Audio Playback Related Errors
 - Error 1.8.1: Wakeup Audio Loading Failed
2. Waste_Classify Module
 - 2.1 Model Loading Related Errors
 - Error 2.1.1: Model File Not Found
 - Error 2.1.2: YOLO Model Initialization Error
 - Error 2.1.3: Model Inference Startup Failed
 - 2.2 Image Detection Related Errors
 - Error 2.2.1: Image Shape Not Initialized
 - Error 2.2.2: Image Save Failed
3. Pcl_Seg Point Cloud Segmentation Module
 - 3.1 Depth Image Processing Errors
 - Error 3.1.1: Invalid ROI Region (Common)
 - Error 3.1.2: No Depth Image
 - Error 3.1.3: No Valid Depth Points In Circular Region
 - Error 3.1.4: Target Point Outside Image Boundary
 - Error 3.1.5: Invalid Radius Value
 - 3.2 TF Transform Related Errors
 - Error 3.2.1: TF Transform Not Available
 - Error 3.2.2: TF Transform Failed
 - 3.3 Point Cloud Processing Related Warnings

Error 3.3.1: ROI Point Cloud Empty	
Error 3.3.2: No Cluster Found	
Error 3.3.3: No Valid Center Data	
4. Super_Tracker Target Tracking Module	
4.1 Tracker Initialization Errors	
Error 4.1.1: Tracker Creation Failed	
Error 4.1.2: Display Window Configuration Conflict	
4.2 Image Frame Processing Warnings	
Error 4.2.1: Received Empty Or Invalid Frame	
4.3 ROI Selection Errors	
Error 4.3.1: Cannot Select ROI On Empty Frame	
Error 4.3.2: Invalid ROI Selected	
Error 4.3.3: Set Track ROI Failed	
Error 4.3.4: Tracking Start Failed	
5. General Troubleshooting Suggestions	
5.1 Log Level Adjustment	
5.2 Common Diagnostic Commands	

1. Multi_Brains_Pre Package

1.1 Configuration File Related Errors

Error 1.1.1: Configuration File Not Found

Error Log:

```
multi_brains_pre_setting.yaml file not found, Please check: whether the file exists;; {config_file}
```

Cause:

- The multi_brains_pre_setting.yaml configuration file path does not exist
- The file has been deleted or moved

Solution:

1. Copy multi_brains_pre_setting.yaml from the attached source code to the specified directory
-

Error 1.1.2: YAML Parsing Failed

Error Log:

```
multi_brains_pre_setting.yaml file YAML parsing failed: {exc}, Please check: whether the file format has been modified
```

Cause:

- YAML file format error (incorrect indentation, unescaped special characters, etc.)

Solution:

1. Modify multi_brains_pre_setting.yaml, refer to the factory file format in the attached source code
-

1.2 ASR Speech Recognition Related Errors

Error 1.2.1: ASR Engine Initialization Failed

Error Log:

```
ASR Engine: {model_name} initialized failed
```

Cause:

- API Key configuration error or invalid
- Model name not supported
- Network issue (online ASR)
- Key file decryption failed (Xunfei ASR)

Solution:

1. Check if `ALIYUN_API_KEY` is correctly configured
 2. Confirm the model name used is in the supported list:
 - paraformer-realtime-v2/v1
 - paraformer-realtime-8k-v2/v1
 - gummy-realtime-v1/gummy-chat-v1
 3. For Xunfei ASR, check the key file path and decryption
 4. Test network connection
-

Error 1.2.2: Speech Recognition Failed

Error Log:

```
asr recognize failure  
online ASR recognition failed: {error_message}
```

Cause:

- API invalid
- Online speech recognition service unavailable

Solution:

1. Check network connection stability
 2. Check if API quota has been exhausted
-

Error 1.2.3: VAD Engine Initialization Failed

Error Log:

```
VadEngine initialization failed  
VadEngine initialization exception: {exc}
```

Cause:

- VAD model file does not exist
- Model path configuration error

- voice_toolbox library not properly installed

Solution:

1. Check if the `silero_vad.int8.onnx` model file has been deleted
 2. Download the VAD model file from the attached source code
-

Error 1.2.4: Audio Stream Open Failed

Error Log:

```
Audio stream open failed: {exc}
```

Cause:

- Microphone device not connected
- Audio device occupied by another program

Solution:

1. Check if the microphone device is properly connected
 2. Use `arecord -l` to view available audio devices
 3. Close other programs occupying the audio device
-

Error 1.2.5: No Speech Detected

Warning Log:

```
No speech detected, timeout exit.
```

Cause:

- Microphone volume too low
- Ambient noise too high

Solution:

1. Increase the microphone input volume
 2. Reduce ambient noise
 3. Speak closer to the microphone
-

Error 1.2.6: Unsupported Language

Warning Log:

```
Unsupported language: {language}, only 'zh' and 'en' are supported
```

Cause:

- Configured unsupported language code
- LANGUAGE parameter set incorrectly

Solution:

1. Change the LANGUAGE parameter to 'zh' (Chinese) or 'en' (English)

2. Check the language setting in the configuration file
`$HOME/M3Pro_ws/src/m3pro_bringup/config/multi_brains_pre_setting.yaml`
-

1.3 TTS Speech Synthesis Related Errors

Error 1.3.1: TTS Engine Initialization Failed

Error Log:

```
Failed to initialize Piper TTS engine: {e}  
Failed to initialize Baidu TTS engine: {e}  
Tongyi TTS_Engine initialized failed  
Baidu TTS_Engine initialized failed  
Xunfei TTS_Engine initialized failed  
Piper TTS_Engine initialized failed
```

Cause:

- TTS model file missing
- API Key configuration error
- Configuration file read failure
- Key file decryption failed (Xunfei TTS)

Solution:

1. Piper TTS:

- Check if the model file exists in the `~/MODELS/tts/` directory
- Confirm both `.onnx` and `.onnx.json` files exist

2. Online TTS (Alibaba Cloud/Baidu/Xunfei):

- Check if the corresponding API Key is correctly configured
- Verify network connection

3. Check if the configuration file format is correct

Error 1.3.2: Speech Synthesis Failed

Error Log:

```
PiperTTS Failed to synthesize text: {e}  
Speech synthesis failed. Check whether the TTS model is available
```

Cause:

- TTS engine not properly initialized?
- Online speech service unavailable?

Solution:

1. Confirm the TTS engine has been properly initialized
 2. Check the specific error message for targeted handling
-

1.4 Voice Wakeup Module Serial Communication Related Errors

Error 1.4.1: Serial Port Open Failed

Error Log:

```
[kws_mic] Failed to open serial {port}: {e}  
Cannot open serial port {port}, please check voice wakeup module connection
```

Cause:

- Voice wakeup serial device does not exist
- Hardware connection issue

Solution:

1. Check if the serial device exists: `ls -l /dev/mic`
 2. Check if the hardware connection is secure
-

Error 1.4.2: Serial Port Reconnection Failed

Warning Log:

```
[kws_mic] Max reconnect attempts ({max_attempts}) reached, giving up  
[kws_mic] Reconnect attempt {attempt} failed: {e}
```

Cause:

- Hardware device disconnected

Solution:

1. Replug the USB device
-

Error 1.4.3: Serial Port Send Failed

Error Log:

```
send fail: {e}
```

Cause:

- Serial connection disconnected
- Buffer overflow
- Voice wakeup device not responding

Solution:

1. Check serial connection status
 2. Restart the program
 3. Check if the hardware device is working properly
-

1.5 ROS Communication Related Errors

Error 1.5.1: Action Server Unavailable

Warning Log:

```
play_audio action server not available, skip wakeup voice  
play_audio action server not available, skip error audio  
play_audio action server not available, skip voice
```

Cause:

- common_service node not started

Solution:

1. Confirm the common_service node has started
 2. Use `ros2 action list` to check available actions
-

Error 1.5.2: Action Goal Rejected

Warning Log:

```
play_audio goal rejected  
TTS play goal rejected
```

Cause:

- Audio file path does not exist

Solution:

1. Check if the audio file exists
 2. Check the specific rejection reason in the logs
-

1.6 Large Model Service Related Errors

Error 1.6.1: Dify Connection Failed

Error Log:

```
Dify LLM connection failed, Please check whether DIY dify has been started, API  
and BASE_URL settings  
Error log: {error_message}  
Dify service connection failed: {status_code}  
Dify service connection failed: {str(e)}
```

Cause:

- Dify service not started
- BASE_URL configuration error
- **API Key invalid** (most common issue, the Dify app API does not match DIFY_API_KEY in the configuration file)

Solution:

1. Confirm the Dify service has started and is running properly
 2. Check the DIFY_BASE_URL configuration (e.g., <http://localhost/v1>)
 3. Verify the DIFY_API_KEY is correct
 4. Use curl to test the connection: `curl -H "Authorization: Bearer {API_KEY}" {BASE_URL}/meta`
-

Error 1.6.2: Model Request Failed

Error Log:

```
The model request failed. Check whether the dify or AI model is normal.  
Error Log: {error_message}
```

Cause:

- Dify service异常
- Model service down
- Model API has no quota
- Timeout

Solution:

1. Check Dify service status
 2. Check Dify service logs
 3. Check API account
 4. Restart the Dify service
-

1.7 Motion Control Related Errors

Error 1.7.1: TF Transform Acquisition Failed

Error Log:

```
Failed to get base_link-roi transform after {max_attempts} attempts, check the  
tf_tree is normal ?
```

Cause:

- TF tree incomplete
- pcl_segment node not publishing TF
- Target tracking lost the target

Solution:

1. Use `ros2 run tf2_tools view_frames` to check the TF tree
 2. Confirm the pcl_segment node is running properly
 3. Ensure the node is tracking the target
-

Error 1.7.2: TF Timestamp Too Old

Warning Log:

```
tf too old for base_link-roi: {age:.3f}s ago (attempt {attempt}/{max_attempts})
```

Cause:

- TF publish frequency too low
- System latency too high, CPU load too high
- Target tracking lost the target

Solution:

1. Increase TF publish frequency
 2. Optimize system performance
 3. Increase the `tf_tolerance` parameter value
 4. Disable unnecessary nodes
-

Error 1.7.3: Joint Values Not Initialized

Warning Log:

```
joint1、joint_2、joint_4 values not initialized, skipping gaze control.
```

Cause:

- Robotic arm query service call failed
- Initialization process not completed

Solution:

1. Check if the robotic arm service is normal
 2. Verify if the arm_util node has started
 3. Check the service call logs
-

1.8 Audio Playback Related Errors

Error 1.8.1: Wakeup Audio Loading Failed

Error Log:

```
Failed to load the wake-up word audio file :{e}
```

Cause:

- Audio file directory does not exist
- File format not supported

Solution:

1. Check the `system_vioce/zh/wakeup` or `system_vioce/en/wakeup` directory
2. Confirm the WAV file exists and the format is correct
3. Check file read permissions
4. Reinstall the multi_brains_pre package

2. Waste_Classify Module

2.1 Model Loading Related Errors

Error 2.1.1: Model File Not Found

Error Log:

```
Model file not found: {model_path}
```

Cause:

- YOLO model path configuration error
- File deleted or moved

Solution:

1. Check the `model_path` parameter configuration
2. Confirm the model file exists
3. Re-download the YOLO model file from the attached source code

Error 2.1.2: YOLO Model Initialization Error

Error Log:

```
YOLO Model initialization error :{e}
```

Cause:

- Factory system Python environment corrupted
- Model file format error
- Insufficient memory

Solution:

1. Check if the model file is in a valid ONNX format
2. Try using CPU mode: set the `device` parameter to "cpu"
3. Check if system memory is sufficient

Error 2.1.3: Model Inference Startup Failed

Error Log:

```
Model inference failed to start: {str(e)}
```

Cause:

- Shared memory creation failed
- Multi-process startup failed
- Insufficient resources

Solution:

1. Check system shared memory limit: `sysctl kernel.shmmax`

2. Check system logs for detailed error information

2.2 Image Detection Related Errors

Error 2.2.1: Image Shape Not Initialized

Error Log:

```
Image shape not initialized, please ensure the camera is running and publishing images
```

Cause:

- Camera not started
- Image topic not published

Solution:

1. Confirm the camera node has started
2. Check if the topic `/camera/color/image_raw` has data: `ros2 topic echo /camera/color/image_raw`
3. Restart the carbase launch file

Error 2.2.2: Image Save Failed

Error Log:

```
Failed to save image: {e}
```

Cause:

- Output directory does not exist
- Insufficient disk space
- Camera node not started

Solution:

1. Check the file path configuration

3. Pcl_Seg Point Cloud Segmentation Module

3.1 Depth Image Processing Errors

Error 3.1.1: Invalid ROI Region (Common)

Warning Log:

```
Invalid ROI Region: [umin vmin umax vmax]
```

Cause:

- Track box coordinates exceed image boundaries
- Track box dimensions are zero or negative

- No depth information in the target region

Solution:

1. Check if the track box coordinates are reasonable
 2. Check in rviz if the target has depth information
-

Error 3.1.2: No Depth Image

Warning Log:

No depth image received.

Cause:

- Depth camera not started
- Topic subscription failed
- Depth image not published

Solution:

1. Confirm the depth camera driver has started
 2. Check if the topic `/camera/depth/image_raw` has data
 3. Verify camera hardware connection
 4. Restart the camera node
-

Error 3.1.3: No Valid Depth Points In Circular Region

Warning Log:

No valid depth points found in the circular region.

Cause:

- Target point too far or too close, no depth information near the target point
- Depth value invalid (0 or 超出閾值)

Solution:

1. Adjust the camera viewing angle and target point position
 2. Check if the depth camera is working properly
 3. Check in rviz if the target has depth information
 4. Reduce ambient lighting
-

Error 3.1.4: Target Point Outside Image Boundary

Warning Log:

The target point is outside the image boundary. Image size: {width}x{height}, Requested: ({x}, {y})

Cause:

- Requested coordinates exceed image range

- Coordinate calculation error
- Image resolution change

Solution:

1. Verify the coordinate calculation logic
 2. Check the image resolution configuration
 3. Add boundary checking
 4. Use valid coordinate values
-

Error 3.1.5: Invalid Radius Value

Warning Log:

```
Invalid radius value.
```

Cause:

- Radius parameter is negative or zero
- Parameter passing error

Solution:

1. Ensure the radius value is positive
 2. Check the service call parameters
 3. Use a reasonable radius value (recommended 5-50 pixels)
-

3.2 TF Transform Related Errors

Error 3.2.1: TF Transform Not Available

Warning Log:

```
TF transform not available between base_link and camera_color_optical_frame.
```

Error Log:

```
Failed to lookup transform: {ex.what()}
```

Cause:

- Missing necessary transform in the TF tree
- Camera coordinate system not configured
- TF publish node not started

Solution:

1. Use `ros2 run tf2_tools view_frames` to check the TF tree
 2. Ensure the camera driver publishes the correct TF
 3. Check if the transform from base_link to camera is correctly defined
-

Error 3.2.2: TF Transform Failed

Error Log:

```
Failed to transform actual_pose: {ex.what()}  
Failed to transform estimate_pose: {ex.what()}  
Failed to transform point: {ex.what()}
```

Cause:

- Coordinate system does not exist
- Transform data异常
- Timestamp issue

Solution:

1. Check if the coordinate system name is correct
 2. Verify TF data validity
 3. Check timestamp synchronization
 4. Use `ros2 run tf2_tools echo` to check TF data
-

3.3 Point Cloud Processing Related Warnings

Error 3.3.1: ROI Point Cloud Empty

Debug Log:

```
ROI cloud is empty, skipping this depthframe
```

Cause:

- No valid depth values within the ROI region
- Distance threshold set too small
- Target超出 distance range

Solution:

1. Increase the `distance_threshold` parameter
 2. Adjust the ROI region size
 3. Ensure the target is within valid distance
 4. Check depth image quality
-

Error 3.3.2: No Cluster Found

Debug Log:

```
No cluster found  
No suitable cluster found
```

Cause:

- Insufficient point cloud density
- Clustering parameters set improperly
- Target too small or too far

Solution:

1. Decrease the `cluster_min_size` parameter
 2. Increase the `cluster_max_size` parameter
 3. Adjust the `ClusterTolerance` parameter
 4. Ensure the target is close enough and clear
-

Error 3.3.3: No Valid Center Data**Warning Log:**

```
GetBoxCenter: no valid data.
```

Cause:

- Depth image not yet processed
- Clustering failed

Solution:

1. Ensure depth image is being processed properly
 2. Check if point cloud segmentation was successful
-

4. Super_Tracker Target Tracking Module

4.1 Tracker Initialization Errors

Error 4.1.1: Tracker Creation Failed**Error Log:**

```
Failed to create tracker: {e.what()}
```

Cause:

- Model file does not exist

Solution:

1. Check the model file path: `$HOME/MODELS/tracker/mixformer_v2_sim.onnx`
 2. Adjust the `num_threads` parameter
-

Error 4.1.2: Display Window Configuration Conflict**Error Log:**

```
display_windows and display_windows_only_track cannot both be true. Setting both to false.
```

Cause:

- Both display parameters set to true simultaneously

- Configuration logic conflict

Solution:

1. Enable only one of the parameters:
 - `display_windows` : Always show window, supports mouse selection
 - `display_windows_only_track` : Show window only during tracking
 2. Modify the configuration file to ensure both are not true at the same time
-

4.2 Image Frame Processing Warnings

Error 4.2.1: Received Empty Or Invalid Frame

Warning Log:

```
Received empty or invalid frame from camera
```

Cause:

- Camera not publishing images properly

Solution:

1. Check the camera node running status
 2. Use `ros2 topic hz /camera/color/image_raw` to check publish frequency
-

4.3 ROI Selection Errors

Error 4.3.1: Cannot Select ROI On Empty Frame

Error Log:

```
Cannot select ROI on empty frame
```

Cause:

- No image frame received yet
- Image frame is empty

Solution:

1. Wait for the camera to publish images normally before selecting ROI
 2. Check camera connection
 3. Confirm the image topic has data
-

Error 4.3.2: Invalid ROI Selected

Warning Log:

```
Invalid ROI selected
```

Cause:

- User cancelled ROI selection
- Selected ROI width or height is zero or negative

Solution:

1. Re-select a valid ROI region
2. Ensure the ROI region has sufficient size (recommended at least 20x20 pixels)
3. Fully框选 the target in the image

Error 4.3.3: Set Track ROI Failed

Error Log:

```
Invalid frame when setting track ROI  
Failed to Set Track ROI {e.what()}
```

Cause:

- Image frame is empty
- Tracker initialization异常
- ROI coordinates invalid

Solution:

1. Ensure a valid image frame is passed
2. Check if ROI coordinates are within the image range
3. Verify tracker status
4. Re-initialize the tracker

Error 4.3.4: Tracking Start Failed

Error Log:

```
Failed to start tracking
```

Cause:

- Current camera frame invalid
- ROI setting failed
- Tracker异常

Solution:

1. Confirm camera image is normal
2. Check ROI coordinate validity
3. Check the specific error log

5. General Troubleshooting Suggestions

5.1 Log Level Adjustment

- If you encounter issues, enable debug log mode and send the complete error output to technical support
- ROS log level: `ros2 launch xxxx --ros-args --log-level debug`

5.2 Common Diagnostic Commands

```
# Check topic list
ros2 topic list

# Check topic data
ros2 topic echo <topic_name>

# Check service list
ros2 service list

# Check Action list
ros2 action list

# Check TF tree
ros2 run tf2_tools view_frames

# Check node status
ros2 node list

# Check logs
ros2 launch xxxx --ros-args --log-level debug
```